

Cognate Effects during Speech Production in Russian-English-German Trilinguals

Lauren Covey and Anastasia Mikheeva

Many studies have examined how bilinguals recognize cognates, words those that share orthographic and phonological form and meaning in two languages (e.g., English *banana*, German *Banane*), as a method to investigate how bilinguals' languages exert influence over one another, even when only one language is used. This research has demonstrated robust *cognate facilitation effects*, such that cognate words are recognized and/or named faster and more accurately than non-cognate words (e.g., Costa et al., 2000; Dijkstra & Van Heuven, 2002; Peeters et al., 2013). In well-designed experiments in which the cognate and non-cognate words have been carefully matched for all other dimensions besides language membership (e.g., frequency, length), cognate facilitation effects are taken to suggest that bilinguals' other language must have played a role in processing, even when they are completing a language-exclusive task in only one of their languages. In other words, there is much evidence that both of bilinguals' languages, target and non-target, are activated during lexical retrieval of a target-language candidate. Thus far, research has primarily focused on bilingual processing, although there is an emerging line of studies showing cognate facilitation effects in trilinguals (e.g., Szubko-Sitarek, 2011; Van Hell & Dijkstra, 2002; Zhu & Mok, 2020). This research provides interesting insights into the structure of the mental lexicon for multilinguals, investigating whether non-selective activation is also in evidence for trilinguals. Trilingual speakers may show crosslinguistic influence from their first language (L1) or second language (L2) when exclusively utilizing their third language (L3); it is also possible that co-activation may emerge in processing any one of their languages (e.g., L1/L3 influence during L2 processing) making it an interesting test case to probe interlingual interaction. In what follows, a brief review of the literature on trilingual cognate effects is provided.

1. Background

Only a handful of studies have thus far examined cognate recognition by trilinguals. For example, Lemöfer et al. (2004) tested Dutch-English-German trilinguals using a lexical decision task in participants' L3 German. Results showed faster reaction times (RTs) for 'double cognate' words which overlapped

* Lauren Covey, Montclair State University, coveyl@montclair.edu. Anastasia Mikheeva, Intento, Inc.

in L1 Dutch and L3 German, as well as a crucial ‘triple cognate’ effect for words which were cognates in all three languages. Similar findings were reported by Szubko-Sitarek (2011) with Polish-German-English trilinguals, reporting faster RTs for triple cognates as compared to non-cognates. A cognate effect was also observed for double cognates (L1 Polish-L3 German). Overall, these findings have been interpreted as support for parallel activation, such that during L3 word retrieval, trilinguals’ non-target languages, L1 and L2, are simultaneously co-activated. This implies that there is non-selective access, such that all of an individual’s known languages may affect word activation and retrieval.

There are several important factors that may affect the degree to which cross-language activation is observed in bilinguals and trilinguals. Language proficiency is one such key variable. Van Hell and Dijkstra (2002) found, for example, that only trilinguals who were highly proficient in their L3 (French) showed a cognate facilitation effect for L1-L3 words during L1 processing. To carefully examine the role of proficiency, as well as to explore effects of language immersion, Poarch and Van Hell (2014) conducted three experiments using a picture naming task: Experiment 1 (German-English-Dutch, L3-immersed), Experiment 2 (Dutch-English-German, L1-immersed) and Experiment 3 (Russian-English-German, L3-immersed). In all three experiments, cognate facilitation effects emerged in both L2 and L3. Importantly, not only did the dominant L1 exert an effect over L2/L3, but triple cognates consistently showed the greatest facilitation, indicating non-dominant languages learned later in life can exert influence on one another. In an analysis collapsing across all three experiments, Poarch and Van Hell found that cognate effects L3 naming (both triple and double) were related to length of L3 immersion and L3 self-assessed proficiency. This suggests that both proficiency and immersion in a language environment play a role in the magnitude of cognate effects for trilinguals.

Another consideration concerns the role of script as well as typological similarity. Poarch and Van Hell’s (2014) third experiment included a group whose L1 Russian had a different script (cyrillic) than their L2 English/L3 German. This investigation was in response to a debate regarding whether cognate facilitation effects only emerge when there is orthographic overlap between the L1 and L2 (Williams, 1994; Bowers et al., 2000) or whether cross-language activation is observed for different script languages (e.g., English/Hebrew, Gollan et al., 1997; Japanese/English, Hoshino & Kroll, 2008). Poarch and Van Hell reported cognate facilitation for Russian-English-German on par with their first two experimental groups, suggesting that language coactivation occurs at the level of phonology during the retrieval process, resulting in cognate effects even for different-script trilinguals.

A recent study by Zhu and Mok (2020) explored typological similarity, testing cognate effects via a lexical decision task with Cantonese-English-German trilinguals. Participants were living in Hong Kong at time of testing, where both Cantonese and English are spoken widely. Their study differed in that ‘triple cognates’ among these languages are not likely to occur, and therefore English-German ‘double’ cognate pairs were utilized. This investigation provides an interesting test case of trilinguals whose L1 is typologically distant from their

L2/L3, and probes the role of the language environment as well. Participants showed facilitation for cognate words in L2 English, as indexed via faster RTs and higher response accuracy. It is particularly interesting to observe cognate facilitation effects from the L3 to the L2, given that the trilinguals were less proficient in L3 German than L2 English. Furthermore, because the L3 was recently learned in adulthood, these results show that the weaker L3 German had influence over stronger L2 English, which was acquired in childhood. Similar facilitation was observed in L3 German (although only in a follow-up experiment). In sum, the authors argue in support of non-selective lexical access during L2 and L3 processing, which was observed for participants immersed in a bilingual L1/L2 environment.

To conclude, several studies have provided evidence of the nonselective nature of trilingual lexical access in same- and different-script trilingual speakers. Additional studies are needed to tease apart the specific contribution of orthography, proficiency, and language environment. Although there is evidence to suggest that cognate effects can emerge in L1-immersion settings (Poarch & Van Hell, 2014; Szubko-Sitarek, 2011) and for trilinguals with different-script L1 (Zhu & Mok, 2020), thus far no study has examined the combination of these factors within one participant group. Furthermore, while there are studies that imply an important role for proficiency, it is common for self-assessed ratings or years of language experience/education to be used in lieu of independent proficiency measures. The current study aims to address these open questions.

2. Current Study

The objective of the current study is to investigate cross-language activation in trilinguals via the cognate facilitation effect, further examining the role of script, proficiency, and language immersion. For this purpose, we conducted a partial replication of Poarch and Van Hell's (2014) Experiment 3 using a picture naming task. The study was completed online via Zoom with Russian-English-German trilinguals who were living in L1-speaking countries at time of testing.

Our primary research question concerned whether we could replicate Poarch and Van Hell's findings in a group of participants with relatively lower L2 English/L3 German proficiency, which was associated with L1-immersion. Poarch and Van Hell showed that despite differences in the script of Russian and English/German, cross-language activation in picture naming did emerge (see also Hoshino & Kroll, 2008), which suggests that we may observe similar facilitation. With regards to proficiency, Van Hell and Dijkstra (2002) argue that a high level of proficiency is required for a weaker language to exert effects on a dominant language (c.f. Zhu & Mok, 2020). Proficiency, assessed via self-ratings and independent measurements, has also been shown to modulate L3 picture naming (Poarch & Van Hell, 2014). We thus included scores from independent measures of L2 and L3 proficiency in our analyses to probe such effects.

Turning now to the role of language exposure and immersion, Poarch & Van Hell (2014; Exp. 3) found that length of L3 immersion was related to the cognate facilitation effect. However, in their second experiment with L1-immersed Dutch-

English-German trilinguals, they did find cognate facilitation effects in evidence, even for a group of participants who had never been immersed in an L3 environment. The participants in the current study are similarly L1-immersed, which suggests that cognate effects may also emerge in our study.

The current study was conducted during the COVID-19 shutdown and the experiment was implemented fully online. Although both naming accuracy and latency were analyzed, the online methodology makes it difficult to interpret naming latency results in comparison with laboratory studies, and we therefore label this investigation as exploratory. However, the goal of this study was to investigate whether cognate facilitation effects emerge in L2 and L3 picture naming in terms of both offline (accuracy) and online (latency) measurements.

3. Methods

3.1. Participants

Twenty trilingual native speakers of Russian who were L2 learners of English and L3 learners of German were recruited for the study (13 female, 6 male, 1 non-binary). All participants resided in Russian-speaking countries (e.g., Russia, Belarus, Ukraine) at time of testing and were recruited online via social media and email. The age of participants ranged from 19 to 42 years ($M=29.9$, $SD=5.5$). Education status varied among participants, with one enrolled in high school, two high school graduates, two 4-year college graduates, and the rest had completed professional degree, with 2 completing a doctorate. As a group, participants had an average of 9.7 years of English instruction ($SD = 4.8$) and 5.2 years of German instruction ($SD=5.0$); English instruction proceeded German instruction for all participants. Time spent in L2- and L3-environments was overall low, with an average 0.5 years ($SD=1.8$) for English and 1.1 years ($SD=1.9$) for German.

Proficiency was assessed via self-ratings on a scale of 1 (very poor) to 5 (very good), as well as via an independent proficiency measure (see Table 1). The LexTALE (Lemhöfer & Broersma, 2012) has been validated as a measure of language proficiency and was administered in English and German. In the task, participants are asked to determine whether a given string is an existing word in the target language. Word frequency varies across items. Well-formed non-words are included such that the final score takes into account the number of correctly identified words and false positives. The task takes approximately 3-5 minutes to administer and consists of 60 items.

Table 1. Mean proficiency scores for L2 English and L3 German

Self-rated (1-5)	L2 English		L3 German	
	<i>Mean (SD)</i>	<i>Range</i>	<i>Mean (SD)</i>	<i>Range</i>
Listening	4.1 (1.2)	2-5	3.4 (0.9)	1-5
Speaking	3.9 (1.2)	2-5	3.1 (0.8)	2-5
Reading	4.4 (0.7)	3-5	3.7 (1.0)	1-5
Writing	4.0 (1.3)	1-5	3.1 (0.9)	2-5
LexTALE (%)	72.0 (12.9)	50.0-95.0	61.9 (10.8)	47.5-88.8

Participants' performance on the two LexTALE tests were not correlated ($r = .25$, $p = .28$) and were quite varied overall, with a large range of scores in evidence for both English and German. A t -test comparing the LexTALE scores revealed significant differences in L2 and L3 proficiency ($t(19) = 3.10$, $p < .01$). This is in line with participants' self-assessments and reflects their overall higher proficiency in L2 English than L3 German.

3.2. Stimuli

Two sets of stimuli were used for the picture naming task, consisting of L2 English and L3 German items from Poarch and Van Hell's (2014) Experiment 3. Five items from the original list were removed due to not being cognates in L2 and L3 (*thumb*, *butterfly*, *tent*) or to repetition (*parrot*, *turtle*).

Stimuli were organized into three conditions: non-cognate words, double cognates, and triple cognates. Different stimuli were used for Experiment 1 (target language L2 English) and Experiment 2 (target language L3 German). Example stimuli are provided in Table 2. Note that 'double cognates' consisted of L1-L2, L2-L3, and L1-L3 cognate pairs. The stimuli were matched by Poarch and Van Hell (2014) on the following characteristics: frequency (CELEX database, Baayen et al., 1993), word length, name agreement, visual complexity (Székely et al., 2004) and imageability (for detailed information see Poarch & Van Hell, 2014, Table 2 pg. 698). A total of 106 black-white drawings were selected from the Multilingual Picture (MultiPic) databank (Duñabeitia et al., 2018).

Table 2. Example stimuli

<i>Experiment 1</i>				
Condition	English (L2)	German (L3)	Russian (L1)	Trials
Triple cognate	<i>banana</i>	<i>Banane</i>	<i>banan</i>	n=18
Double cognate	<i>apple</i>	<i>Apfel</i>	<i>yablako</i>	n=17
Non-cognate	<i>airplane</i>	<i>Flugzeug</i>	<i>samolet</i>	n=19
<i>Experiment 2</i>				
Condition	German (L3)	English (L2)	Russian (L1)	Trials
Triple cognate	<i>Gitarre</i>	<i>guitar</i>	<i>gitarra</i>	n=18
Double cognate	<i>Esel</i>	<i>donkey</i>	<i>osel</i>	n=18
Non-cognate	<i>Tasse</i>	<i>cup</i>	<i>tschashka</i>	n=18

3.3. Procedure

The experiment was administered individually in one online session conducted with the software Zoom, divided into two parts. All participants took the English tasks before the German tasks. The English LexTALE was administered first, which took approximately 5 minutes, and then participants completed Experiment 1, the English picture naming task. After a short break, the German LexTALE was administered followed by Experiment 2, the German picture naming task. The entire Zoom session took about 30-40 minutes.

The picture naming tasks were run by the experimenter, who shared her screen over Zoom and showed a PowerPoint presentation. The experimenter (second author) is trilingual in Russian, English, and German, and conducted the two parts of the experimental session in the respective target language. All instructions were also presented in the target language.

In the task, participants were shown black-and-white line drawings of common objects and were asked to name each picture in the target language as quickly as they could. The picture naming task included 10 practice trials to familiarize participants with the experiment. The task was recorded via Zoom and downloaded in .m4a format, converted into .wav for analysis in Praat.

Each experimental trial was preceded by a fixation cross displayed for 1000 ms and followed by the target picture for 8000 ms, or until a response was given. The PowerPoint slide included the automatic presentation of an auditory cue (beep) that was co-presented along with the target image; in this way, response latency could be measured from the offset of the cue to the onset of naming. Stimuli were presented in a pseudorandom fixed order with the restrictions that no more than three cognates or non-cognates with an identical initial phoneme were displayed in a row. Two lists with different item presentation orders were created; lists were presented in cross-balanced order across participants.

4. Results

4.1. Naming accuracy

To measure naming accuracy, trial responses were coded as 0 (incorrect) or 1 (correct). To investigate if naming accuracy was influenced by language or experimental conditions, a generalized linear model with a binomial distribution was fit, with Accuracy (0, 1) as the dependent variable. Language (L2, L3) and Condition (non-cognate, double cognate, triple cognate) were included as fixed effects, and the model included random intercepts for Participant and Item. The full model, which included the two fixed effects and interaction term Language $\times$ Condition, was simplified stepwise. The interaction term was removed during model fitting; the best-fitting final model is represented below:

$$\text{Accuracy} \sim \text{Condition} + \text{Language} + (1|\text{Participant}) + (1|\text{Item})$$

Results from the model indicated that there was a main effect of Language, such that participants were overall more accurate naming in L2 English compared to L3 German ($\beta = -1.43$, $SE = 0.12$, $t = -12.18$, $p < .001$). In addition, Condition influenced accuracy. Participants were overall most accurate naming Triple cognates (Non-cognate vs. Triple cognate: $\beta = 2.12$, $SE = 0.16$, $t = 13.41$, $p < .001$; Double vs. Triple cognate: $\beta = 1.45$, $SE = 0.16$, $t = 9.06$, $p < .001$), and were also more accurate naming Double cognates as compared to Non-cognates ($\beta = 0.67$, $SE = 0.13$, $t = 5.32$, $p < .001$). The lack of interaction effect (removed during model fitting) indicates that the role of Condition was the same in L2 English and L3 German, with the highest accuracy in the Triple Cognate condition, followed

by the Double Cognate condition, and the lowest accuracy in the Non-cognate condition. This is shown in Figure 1.

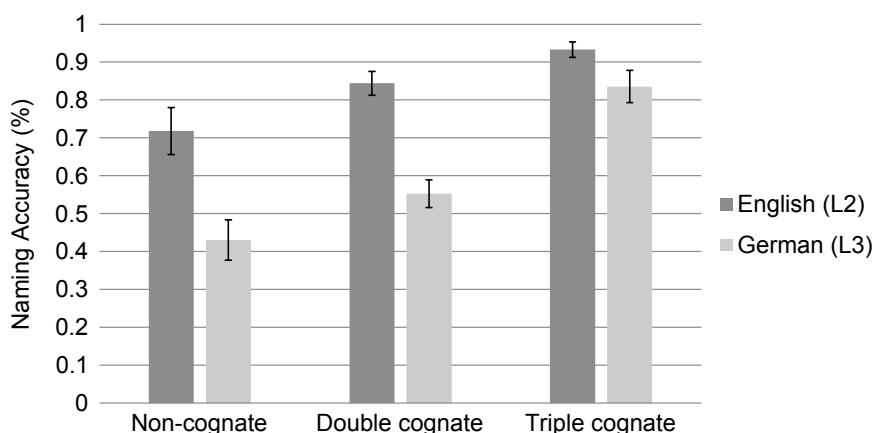


Figure 1. Accuracy rates for L2 and L3 picture naming.

To investigate whether proficiency modulated accuracy, we followed Poarch and Van Hell (2014) and conducted a correlational analysis using LexTALE scores and cognate effect magnitudes, calculated by subtracting non-cognate accuracy rates from triple or double cognate accuracy. For L2 English, we found that English LexTALE scores were significantly correlated with triple cognate effects ($r = -.48, p < .05$) and double cognate effects ($r = -.57, p < .01$). This was also in evidence in L3, where German LexTALE scores were related to double cognate magnitude ($r = -.54, p < .05$). These negative correlations indicate that as proficiency increases, the magnitude of the cognate effect size decreases.

To summarize, we found robust cognate facilitation effects for both L2 English and L3 German, such that participants were most accurate in naming triple cognate words, followed by double cognates, with non-cognates showing the lowest accuracy rates. Participants were overall more accurate in L2 naming, which was expected given their higher proficiency in the L2. Proficiency was significantly related to the magnitude of cognate effects in both languages, such that increased L2/L3 proficiency was associated with reduced cognate effects.

4.2. Naming latency

Latency was analyzed in Praat, measured from the offset of the auditory cue to the onset of naming. Only accurate trials were included in the latency analysis (84% of total trials). Trials were eliminated as outliers if latency was below 200 ms or longer than 2.5 SD above the participant's overall mean. We also followed Poarch and Van Hell (2014) in using the following data trimming procedures to prevent low accuracy rates from biasing the results: participants with less than 40% overall accuracy and items below 25% accuracy excluded from analysis.

In total, one participant was removed from analysis for each experiment and 19.2% of total trials were excluded from analysis. This differed depending on experiment and condition, with 104 trials excluded from L2 English (19.4% non-cognates, 12.9% double cognates, 5.9% triple cognates) and 168 trials excluded from L3 German naming (29.0% non-cognates, 22.1% double cognates, 10.7% triple cognates). Given the overall number of trials excluded, it is important to highlight the latency analysis as *exploratory*, especially given the nature of the data collection, which took place over the Internet and not in a laboratory setting.

Linear mixed-effects models were run for the L2 English and L3 German data separately, with the maximal random effect structure justified by the data (Participant, Item). For both models, the effect of Condition was not significant: L2 English (Non-cognate vs. Double cognate: $\beta = 81.23$, $SE = 102.59$, $t = 0.792$, $p = .432$; Non-cognate vs. Triple cognate: $\beta = 40.48$, $SE = 100.16$, $t = 0.404$, $p = .688$) and L3 German (Non-cognate vs. Double cognate: $\beta = 121.56$, $SE = 163.14$, $t = 0.745$, $p = .461$; Non-cognate vs. Triple cognate: $\beta = 242.19$, $SE = 144.62$, $t = 1.675$, $p = .101$). This lack of main effect indicates that latencies across the three conditions did not differ significantly (Figure 2).

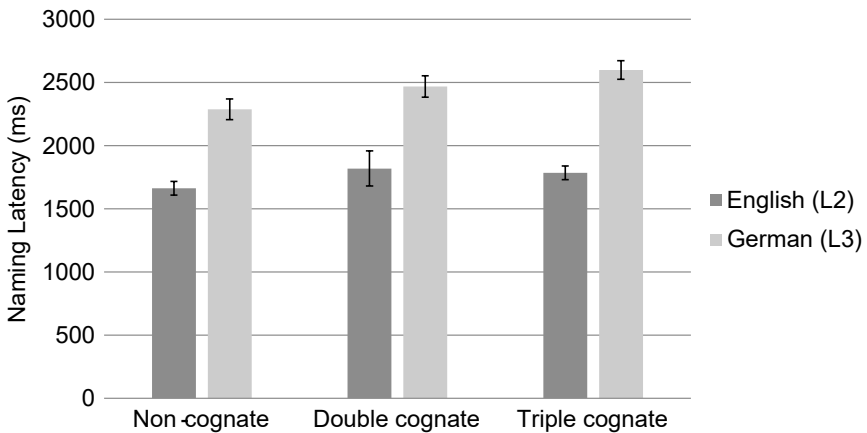


Figure 2. Naming latency for L2 and L3 picture naming.

Although no differences across conditions emerged at the group level, we wanted to investigate whether there were individual differences in naming latency related to proficiency, which was quite variable in our sample. To address this, we conducted a follow-up analysis that included LexTALE scores.

English proficiency scores were included in the L2 English naming latency model, and German scores were used in the L3 latency model. In the L2 model, there was a main effect of Proficiency ($\beta = -3840.0$, $SE = 1066.8$, $t = -3.600$, $p < .01$), which indicated that participants who were more proficient in English were overall faster to name pictures in L2 English. However, LexTALE scores did not interact with Condition, indicating that proficiency modulated naming latency across all conditions similarly.

In the German model, a main effect of Proficiency also emerged ($\beta = -2754.1$, $SE = 1033.3$, $t = -2.655$, $p < .05$). Interestingly, when LexTALE scores were included in the model, a marginally significant main effect of Condition emerged (Non-Cognate vs. Double cognate: $\beta = -1291.4$, $SE = 683.4$, $t = -1.890$, $p = .0594$). An interaction was also in evidence between LexTALE scores and Condition, at the level of Double Cognate ($\beta = 2260.1$, $SE = 1027.1$, $t = 2.200$, $p < .05$). Together, these results indicate that learners with better German proficiency showed significantly faster naming latencies for double cognates over non-cognates.

To illustrate this effect, we calculated an effect size measurement, subtracting non-cognate naming latencies from double cognate latencies for each participant. Figure 3 provides a visualization of the relationship between the double cognate latency effect and German LexTALE scores.

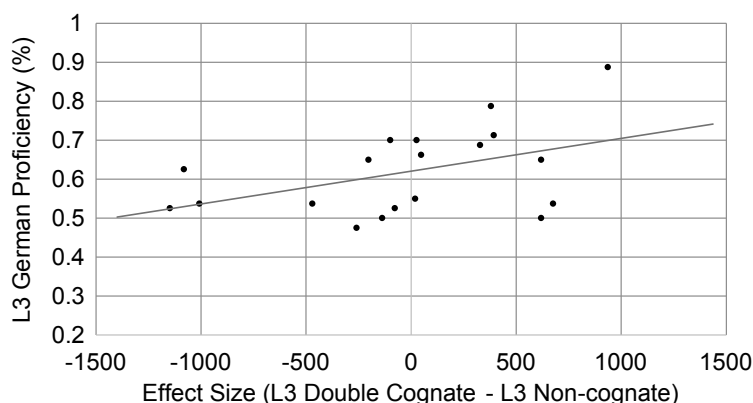


Figure 3. Relationship between double cognate latency effect and German proficiency scores for L3 picture naming.

5. Discussion

Considering first the results for naming accuracy, we found that participants were significantly more accurate in naming pictures in L2 English, which was in line with their overall higher proficiency in the L2. Crucially, in both L2 and L3 naming, participants demonstrated a clear cognate facilitation effect mirroring Poarch & Van Hell's (2014) results. Highest accuracy was observed for triple cognates, over both double and non-cognates. A double cognate effect was also observed, with higher accuracy for double over non-cognates. Thus, at the behavioral level, there was facilitation for cognates in general over non-cognates.

That we were able to largely replicate Poarch and Van Hell's (2014) results with a group of L1-immersed participants is significant. Results suggest that cross-language influence occurred in both L2 and L3 naming, which is also noteworthy given the lower L3 proficiency of our participants. In the original study, Russian trilinguals were matched on their L2 English and L3 German

proficiency, whereas our participants were significantly more proficient in L2 than L3 and had overall low L3 immersion (with L1-immersion at time of testing). Given that participants showed the best performance on triple cognates across experiments, this suggests that the combined influence of all languages impacts naming accuracy in both of participants' non-dominant languages. Although we cannot isolate the source of 'double cognate' effects, given that this condition was comprised of three types of cognates (L1/L2, L1/L3, L2/L3), our results indicate that overlap with all three languages leads to improved performance above and beyond 'simple' overlap from the L1 alone (i.e., triple cognates were named more accurately than even double cognates). These findings are in line with studies which find cross-language influence even for L1-immersed participants (e.g., Lemhöfer et al., 2004; Poarch & Van Hell, 2014 Exp. 2; Szubko-Sitarek, 2011) and for different-script trilinguals (e.g., Poarch & Van Hell, 2014 Exp. 3; see also Hoshino & Kroll, 2008).

With respect to the role of proficiency, we found evidence to suggest that increased proficiency was associated with decreased magnitude of cognate facilitation effects. This pattern is in line with Poarch and Van Hell's (2014) findings, although it was not directly interpreted in their study. The negative correlation was at first surprising to us, but overall, we take reduced cognate effects as being linked to *increased* performance on non-cognate items. In other words, higher proficiency participants show better performance across conditions, whereas lower proficiency is associated with higher accuracy for words that are shared across languages (double and triple cognates). This is roughly in line with Kroll et al.'s (2005) review paper, which showed that as proficiency increased in bilinguals' L2, the magnitude of cognate effects becomes more similar between their L1 and L2. In our study, proficiency modulated effect size in both the L2 and L3 similarly, both of which differed in script from participants' L1.

Turning to the results of naming latency, we first wish to highlight the exploratory nature of this analysis, the results of which should not be directly compared to laboratory studies. The online nature of the experiment means that latency measurements are subject to internet speed of each participant and the online software (Zoom), although we went forward with the analysis under the notion that internet speed was relatively consistent within a given participant's session. In addition, it is noteworthy that participants' latency in general was longer in our study than in Poarch and Van Hell (2014), in some cases as much as 500 – 1,000 ms longer on average (e.g., in L3 naming). This may be a reflection of the methodology or the overall lower L3 proficiency of participants; we hope to conduct future in-person studies to address these open questions.

In contrast to Poarch and Van Hell (2014), we did not observe a cognate facilitation effect in naming latency. The only factor that modulated latency was language, with participants overall faster to name pictures in L2 English. To better understand whether the lack of cognate effects were derived from proficiency-related factors, we conducted a follow-up analysis that included LexTALE scores in the LME models. This analysis revealed that proficiency scores significantly modulated naming latency, but only in the L3. Specifically, higher German proficiency was associated with a larger double cognate effect, i.e., greater

facilitation for naming double cognate words in German. This finding is in line with proposals that argue for the importance of proficiency (e.g., Costa et al., 2000; Van Hell & Dijkstra, 2002), although several open questions remain.

Our first question concerned why this relationship emerged for double cognates in particular, as compared to triple cognates. One possibility raised by Poarch and Van Hell (2014) is that for L1-immersed trilinguals, it may be the case that the mapping of triple cognates in the lexicon is somewhat less extensive and integrated to a lesser degree than for trilinguals immersed in an L2- or L3-environment. This pattern could also suggest a somewhat weaker role for L1 activation, which would be best supported by greater facilitation for triple cognates, which share phonological and semantic overlap across all three languages. However, because the double cognate condition included three types of cognate items, it is difficult to disentangle the specific contribution of the L1. Future trilingual studies should aim to include distinct double cognate conditions to better understand this double cognate facilitation (e.g., Zhu & Mok, 2020).

Another open question concerned why German proficiency modulated cognate facilitation effects in L3 naming, but English proficiency did not modulate L2 naming latency, despite the fact that learners were more proficient in English. We consider several possibilities. First, as the most recently acquired language, L3 German may be subject to increased cross-language activation from more dominant L1/L2. This is in line with Kroll et al.'s (2005) analysis, who found that L1 influence on L2 remained stronger than L2 on L1 effects, and is also representative of Zhu and Mok's (2020) findings that lexical processing in a lower proficiency L3 is subject to multiple contextual factors. Second, because learners were actively enrolled in (online) German courses at time of testing, it may be the case that the active use and routine exposure to the L3 led to greater variability in naming latency that was associated with individual differences in proficiency. Finally, because L3 picture naming was always completed in the second block after the English tasks, it is possible that additional experimental training may have aided some participants, or potentially resulted in cognitive fatigue. It is difficult to completely rule out several potential factors, and we look forward to additional studies that aim to tease apart the unique contribution of proficiency, language exposure, and the role of the L1.

6. Conclusions

The current study aimed to replicate findings of Poarch and Van Hell (2014; Experiment 3) by extending their methodology to an online setting with L1-immersed trilingual participants. Results demonstrated cognate facilitation effects in both L2 and L3 naming accuracy, replicating previous research and revealing a crucial role for proficiency. Behavioral findings provided evidence that it is possible for to observe cross-language activation even in a relatively low proficient L3, and for different-script trilinguals who are not immersed in an L2 or L3 environment. In an exploratory analysis examining naming latency, we did not observe cognate effects, although there are several reasons this may have occurred, including factors related to the online data collection. We hope that

this study provides a springboard for future studies which may carefully disentangle the interaction(s) between proficiency, language immersion, and language script in trilingual cognate processing.

References

- Baayen, Harald R., Piepenbrock, Richard, & Van Rijn, Hedderik. (1993). The CELEX lexical database. Philadelphia, PA: Linguistic Data Consortium, University of Pennsylvania.
- Bowers, Jeffrey S., Mimouni, Zohra, & Argun, M. (2000). Orthography plays a critical role in cognate priming: Evidence from French/English and Arabic/French cognates. *Memory & Cognition*, 28(8), 1289-1296.
- Costa, Albert, Caramazza, Alfonso & Sebastian-Galles, Nuria (2000). The cognate facilitation effect: implications for models of lexical access. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 26(5), 1283.
- Dijkstra, Ton & Van Heuven, Walter J. B. (2002). The architecture of the bilingual word recognition system: From identification to decision. *Bilingualism: Language and Cognition*, 5(3), 175-197.
- Duñabeitia, Jon Andoni, Crepaldi, Davide, Meyer, Antje, New, Boris, Pliatsikas, Christos, Smolka, Eva & Brysbaert, Marc (2018). MultiPic: A standardized set of 750 drawings with norms for six European languages. *Quarterly Journal of Experimental Psychology*, 71(4), 808- 816.
- Gollan, Tamar H., Forster, Kenneth I., & Frost, Ram. (1997). Translation priming with different scripts: Masked priming with cognates and noncognates in Hebrew-English bilinguals. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 23(5), 1122-1139.
- Hoshino, Noriko & Kroll, Judith (2008). Cognate effects in picture naming: Does cross-language activation survive a change of script?. *Cognition*, 106(1), 501-511.
- Kroll, Judith, Sumutka, Bianca M., & Schwartz, Ana I. (2005). A cognitive view of the bilingual lexicon: Reading and speaking words in two languages. *International Journal of Bilingualism*, 9, 27-48.
- Lemhöfer, Kristin & Broersma, Mirjam (2012). Introducing LexTALE: A quick and valid Lexical Test for Advanced Learners of English. *Behavior Research Methods*, 44, 325-343.
- Lemhöfer, Kristin, Dijkstra, Ton & Michel, Marije (2004). Three languages, one ECHO: Cognate effects in trilingual word recognition. *Language and Cognitive Processes*, 19(5), 585-611.
- Peeters, David, Dijkstra, Ton & Grainger, Jonathan (2013). The representation and processing of identical cognates by late bilinguals: RT and ERP effects. *Journal of Memory and Language*, 68(4), 315-332.
- Poarch, Gregory & Van Hell, Janet (2014). Cross-language activation in same-script and different-script trilinguals. *International Journal of Bilingualism*, 18(6), 693-716.
- Szekely, Anna, Jacobsen, Thomas, D'Amico, Simona, Devescovi, Antonella, Andonova, Elena, Herron, Daniel, Lu, Ching Ching, Pechmann, Thomas, Pléh, Csaba, Wicha, Nicole, Federmeier, Kara, Gerdjikova, Irina, Gutierrez, Gabriel, Hung, Daisy, Hsu, Jeanne, Gowri, Iyer, Kohnert, Kathryn, Mehotcheva, Teodora, Orozco-Figuera, Araceli, Tzeng, Angela, Tzeng, Ovid, Arévalo, Analía, Vargha, Andras, Butler, Andrew C., Buffington, Robert, & Bates, Elizabeth (2004). A new on-line resource for psycholinguistic studies. *Journal of Memory and Language*, 51(2), 247-250.

- Szubko-Sitarek, Weronika (2011). Cognate facilitation effects in trilingual word recognition. *Studies in Second Language Learning and Teaching*, 1(2), 189-208.
- Van Hell, Janet & Dijkstra, Ton (2002). Foreign language knowledge can influence native language performance in exclusively native contexts. *Psychonomic Bulletin & Review*, 9(4), 780-789.
- Williams, Edwin. (1994). Remarks on lexical knowledge. *Lingua*, 92, 7-34.
- Zhu, Yanjiao & Mok, Peggy Pik Ki (2020). Visual recognition of cognates and interlingual homographs in two non-native languages: Evidence from Asian adult trilinguals. *Linguistic Approaches to Bilingualism*, 10(4), 441-470.

Proceedings of the 47th annual Boston University Conference on Language Development

edited by Paris Gappmayr
and Jackson Kellogg

Cascadilla Press Somerville, MA 2023

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ISSN 1080-692X
ISBN 978-1-57473-087-6 (2 volume set, paperback)

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